

Total No. of Questions : 10]

SEAT No. :

P2917

[Total No. of Pages : 4

[5669]-506

T.E. (Civil)

ADVANCED SURVEYING

(2015 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No. 1 or Q. No. 2, Q.No. 3 or Q.No. 4, Q.No. 5 or Q.No. 6., Q. No. 7 or Q.No. 8, Q.No. 9 or Q.No. 10.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) The triangulation stations A and B 50KM apart are having elevations 243m and 260m respectively. The intervening ground may be assumed to have a uniform elevation, of 216m. Find the minimum height of the signal required at B, so that the line of sight may not pass the near the ground than 2.4m. [6]

b) Explain the three point problem in hydrographic surveying and list out the methods for solving it. [4]

OR

Q2) a) Explain the sources of errors in SBPS positioning. [6]

b) Explain graphical method for solution of three point problem in hydrographic surveying. [4]

Q3) a) Define Remote Sensing and explain use of electromagnetic spectrum in remote sensing. [6]

b) What do you mean by sounding? Explain how soundings are measured. [4]

OR

P.T.O.

- Q4) a)** What is GIS? Enlist and explain the components of GIS. [6]
- b) State different methods of locating soundings used in hydrographic surveying. Explain any one in detail. [4]
- Q5) a)** Explain different sources of errors. [5]
- b) Define : [8]
- Accuracy,
 - Conditioned and normal equation,
 - Observation equation, and
 - Weight
- c) The angles of a spherical triangle PQR were observed as follows : [5]
- $P = 490^{\circ} 11' 13.2''$ weight=1
- $Q = 74^{\circ} 26' 34.8''$ weight=3
- $R = 56^{\circ} 22' 17''$ weight=2
- If the area of the above spherical triangle is 1764 sq. km, adjust the angles of triangle. Take spherical excess equal to 1" for every 196 sq. km area.
- OR
- Q6) a)** Explain the process of adjustment in case of geodetic quadrilateral with no station at the intersection of diagonals. [5]
- b) Determine the most probable values of the angles of a triangle ABC using method of correlates, given by the following data : [8]
- $A = 62^{\circ} 14' 12''$ weight=1
- $B = 48^{\circ} 12' 14''$ weight=3
- $C = 69^{\circ} 33' 28''$ weight=2
- c) What do you mean by spherical excess? Explain the procedure of determining the sides of spherical triangle. [5]

- Q7) a)** Define relief displacement. Derive an expression for the same. [5]
- b)** Define : [5]
- i) Isocentre,
 - ii) Exposure Station,
 - iii) Flying height and
 - iv) Tilt and tip
- c)** The scale of an aerial photograph is 1 cm = 160 m & the size of the photograph is 20 cm × 20 cm. If the longitudinal overlap is 65 % and side overlap is 35 %, determine the number of photographs required to cover an area of 232 sq. Km. [6]

OR

- Q8) a)** What are the different types of aerial photograph? Compare a map with aerial photograph. [5]
- b)** Explain the procedure of determining the minimum number of aerial photographs required to cover a given area. [5]
- c)** Determine the minimum number of aerial photographs required to cover an area of 40 km X 30 km, with the following details : [6]

Size of an aerial photograph = 23 cm X 23 cm

Scale of aerial photograph: 1 cm = 150 m

Longitudinal overlap = 60 %

Side overlap = 30 %

- Q9) a)** Derive an expression for the difference of level between two points A and B distance D apart, with the vertical angle as angle of elevation from A to B. The height of the instrument at A and that of the signal at B are equal. [8]
- b)** Explain the procedure of marking the alignment of the tunnel on the surface of the ground and transferring the alignment underground. [8]

OR

- Q10)a)** It is required to determine the elevation of a station O. Observations were made to three stations A, B and C already fixed and of known elevations. The following data was recorded. **[8]**

Inst. Stn	Stn obs.	Ht. of inst	Distance (m)	Ht. of signal	Vertical angle
O	A	1.50	3600	5.6	1°1' 20"
	B		4700	4.1	-53' 00"
	C		5000	4.9	-34' 10"

- b) Describe the procedure for determining centre line length of bridge and procedure for location of bridge piers while setting out a bridge. **[8]**

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